

Name: _____

FRACTIONS (SECONDARY), (MULTIPLYING FRACTIONS)

Date: _____

LO: I can multiply fractions and mixed numbers, simplifying my answers.

Multiplying fractions

To **multiply fractions**, multiply the **numerators** together and the **denominators** together. Simplify at the end (or cancel first).

Tip: Cross-cancel before multiplying to keep numbers small. Convert mixed numbers to improper fractions first.

Key method

●	$\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$	multiply across
●	$\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$	cancel 3s first
●	$1\frac{1}{2} \times \frac{2}{3} = \frac{3}{2} \times \frac{2}{3} = 1$	convert then multiply
●	$\frac{2}{3} \times \frac{4}{5} = \frac{6}{8}$	added instead of multiplied
●	$\frac{2}{3} \times \frac{4}{5} = \frac{8}{5}$	only multiplied tops

Key idea

Multiply tops together, multiply bottoms together. Cancel common factors for simpler working.

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Straight across: top × top, bottom × bottom

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – simple multiplication

Calculate $\frac{2}{5} \times \frac{3}{4}$

Multiply numerators: $2 \times 3 = 6$

Multiply denominators: $5 \times 4 = 20$

$$\frac{6}{20} = \frac{3}{10}$$

$$= \frac{3}{10}$$

Simplify by dividing top and bottom by 2.

Example 2 – cancelling first

Calculate $\frac{3}{8} \times \frac{4}{9}$

Cancel: 3 and 9 share factor 3; 4 and 8 share factor 4

$$\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

$$= \frac{1}{6}$$

Cross-cancel before multiplying keeps numbers small.

Example 3 – fraction of a whole number

Calculate $\frac{3}{4} \times 20$

Write 20 as $\frac{20}{1}$

Cancel: 20 and 4 share factor 4 $\frac{3}{1}$

$$\times \frac{5}{1} = 15$$

$$= 15$$

$\frac{3}{4}$ of 20: divide by 4 (=5), multiply by 3 (=15).

Example 4 – mixed numbers

Calculate $1\frac{2}{3} \times 2\frac{1}{4}$

Convert: $\frac{5}{3} \times \frac{9}{4}$

Cancel: 9 and 3 share 3 $\frac{5}{1} \times \frac{3}{4}$

$$= \frac{15}{4} = 3\frac{3}{4}$$

$$= 3\frac{3}{4}$$

Always convert mixed numbers to improper fractions before multiplying.

YOUR TURN – PRACTICE (deliberate practice)

1. Simple fraction multiplication

Multiply and simplify.

- a) $\frac{1}{2} \times \frac{3}{4}$
- b) $\frac{2}{3} \times \frac{1}{5}$
- c) $\frac{3}{4} \times \frac{2}{3}$
- d) $\frac{5}{6} \times \frac{3}{10}$

2. Fraction of a whole number

Find each fraction of the number.

- a) $\frac{1}{3} \times 12$
- b) $\frac{2}{5} \times 30$
- c) $\frac{3}{4} \times 24$
- d) $\frac{5}{8} \times 40$

3. Cross-cancelling

Cancel common factors before multiplying.

- a) $\frac{4}{9} \times \frac{3}{8}$
- b) $\frac{5}{12} \times \frac{6}{10}$
- c) $\frac{7}{10} \times \frac{5}{14}$
- d) $\frac{8}{15} \times \frac{5}{4}$

4. Mixed numbers

Convert to improper fractions first.

- a) $1\frac{1}{2} \times \frac{2}{3}$
- b) $2\frac{1}{4} \times \frac{2}{3}$
- c) $1\frac{1}{3} \times 1\frac{1}{2}$
- d) $3\frac{1}{3} \times \frac{3}{5}$

5. Mixed multiplication

Calculate. Give answers in simplest form.

- a) $\frac{2}{7} \times \frac{7}{8}$
- b) $4 \times \frac{3}{8}$
- c) $\frac{9}{16} \times \frac{4}{3}$
- d) $2\frac{1}{2} \times 1\frac{3}{5}$
- e) $\frac{5}{6} \times \frac{2}{5} \times \frac{3}{4}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$ ☐

Correct answer: _____

Reason: _____

b) Multiply tops and multiply bottoms ☐

Correct answer: _____

Reason: _____

c) $\frac{2}{3} \times \frac{4}{5} = \frac{6}{8}$ ☐

Correct answer: _____

Reason: _____

d) You need a common denominator to multiply ☐

Correct answer: _____

Reason: _____

e) Cross-cancel before multiplying to simplify ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A rectangle has width $\frac{3}{4}$ m and length $1\frac{2}{3}$ m. Find its area. Then find the area of a square with side $\frac{5}{6}$ m.

Expression: _____

Simplified: _____

b) Show that $\frac{a}{b} \times \frac{b}{a} = 1$ for any non-zero a and b. What is the name for $\frac{b}{a}$ in relation to $\frac{a}{b}$? Use this to explain why dividing by a fraction is the same as multiplying by its reciprocal.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

